

COAR UPPER EXTREMITY OMM Handout- Diagnosis and Treatment

ABC's

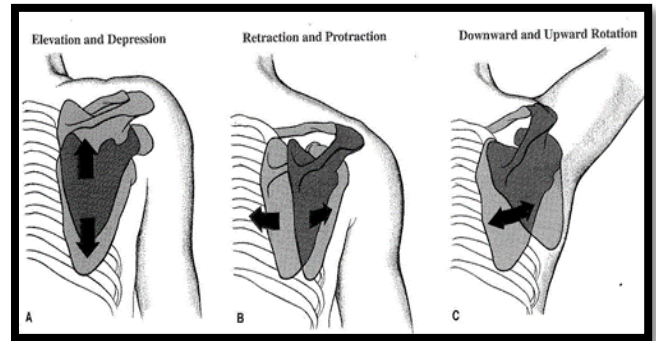
When considering the upper extremity be sure that you consider the ABC's.

- **Autonomics**
 - T1-T5
- **Biomechanics**
 - Gross range of motion
 - Glides and slides of each joint
 - Upper crossed syndrome considerations
- **Circulation**
 - thoracic inlet, diaphragm

DIAGNOSIS:

Scapulothoracic Joint:

- a. Observe for static positional asymmetry/deformity(winging)
- b. Palpate
- c. Ortho component-Active Range of Motion (AROM) testing for instability/gross motion restriction
- d. 3 Motions:
 1. Rotation about an A-P axis – Use to lateralize to most restricted side
 - a. Created by Shoulder Abduction:('Normal ROM'=180 Deg.) palms facing out.
 - b. Observe & palpate for asymmetry & quality of motion. The side that moves first is considered to be the most restricted- as described in prior labs. Observe all the way to the fingers for static asymmetry
 - i. Superior (aka upward) rotation=glenoid fossa facing superior-Abduct arm
 - ii. Inferior (aka downward) rotation=glenoid fossa facing inferior- Adduct arm
 2. Elevation/Depression (Shrug shoulders, reach towards floor)
 3. Protraction/Retraction (move shoulders anteriorly & posteriorly toward midline of body)



Glenohumeral Joint:

Passive ROM- minor motions-Short-lever

1. Stabilize Glenoid fossa by contacting scapular spine & clavicle/acromion
2. Grasp proximal Humerus
3. Ortho component-Motion test Humeral head gently but firmly to end ROM of joint capsule, with respect to Glenoid fossa
 - a. Superior/Inferior Glide
 - feel for instability (Sulcus sign) during **gross** Inferior Glide
 - b. Anterior/Posterior Glide
 - feel for instability during **gross** A-P Glides
 - c. Is gross gliding the same as the gliding used to assess somatic dysfunction?



AC Joint:

1. Patient seated- operator behind the patient- one hand palpates the superior aspect of the AC- the other grasps the forearm

2. Operator, while monitoring at the AC Joint:
 - a. Adduction causes superior glide
 - b. Abduction causes inferior glide
 - c. Internal rotation causes anterior glide
 - d. External rotation causes posterior glide
3. Restriction in symmetric motion denotes somatic dysfunction

SC Joint:

1. Observe for static positional asymmetry /deformity
2. Palpate SC joints & note Asymmetry of position, Tissue Texture, and Tenderness
3. Spring SC Joints to help lateralize: GENTLY spring proximal Clavicles inferiorly, superiorly and posteriorly (Segmentally diagnose and treat side with less spring first)
 - a. Inferior/Superior glide (Have patient shrug shoulders, then reach towards their feet)
 - b. Posterior/Anterior glide (Shoulder Flexion to 90 deg. then reach forward) Protraction about 2-3 inches, followed by Retraction)
 - c. Anterior/Posterior rotation (Abduct arms to 90 deg., Internally rotate humerus until clavicle anteriorly rotates, Externally rotate humerus until clavicle posteriorly rotates -with elbows flexed or straight)

Abducted/Adducted Elbow:

1. Patient is supine and shoulder abducted to 45 degrees. Place the hand midway between supination and pronation (you can then pin it at your side). Grasp the elbow with both hands, induce slight flexion
2. Introduce a very slight medial and lateral motion and observe for ease and tension paying close attention to anatomic barriers.

Radial head:

1. Use Handshake grip
2. Other hand fingers or thumb monitors radial head for response to:
 - a. -Supination-radial head glides anteriorly
 - b. Pronation- radial head glides posteriorly

Interosseous membrane forearm

1. Palpate Interosseous membrane for region of greatest Tissue tension / tenderness
 2. Start proximally on posterior(dorsal) aspect of forearm just distal to Radial head and move distally. once tissue thins out you can palpate either anterior or posterior aspects of distal forearm and reach the IOM
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TREATMENT:

Autonomics and Circulation

Soft tissue to the Thorax/Lumbar

Rib Raising - seated or supine

Thoracic HVLA - Texas Twist (flexed only)

Diaphragm MFR

Thoracic inlet MFR

Biomechanics

Scapulothoracic MFR/ME (Scapular lift)

1. Patient is lateral recumbent, operator faces the patient. Internally rotate their arm and place it behind their back.
2. Operators cephalad hand stabilizes the shoulder and caudad hand grasps the medial border of the scapula
3. Lifting the scapula away from the rib cage, you can treat myofascially or engage restrictive barriers with muscle energy. Respiratory cooperation can also be used.

Seven Stages of Spencer

The patient is side-lying- utilize muscle energy principles- move into the motion barrier- have the patient isometrically contract in the other direction, relaxation, engage the next barrier

1. Extension
2. Flexion
3. Circumduction with compression
4. Circumduction with traction
5. Abduction
6. Internal rotation
7. Joint pump

ME for the AC Joint

1. Patient is seated- stand in the same position as when diagnosing (as listed above)
2. Position the patient into the restrictive barriers (restricted glides)
3. Have the patient utilize ME pushing in the direction opposite of the restrictive barrier
4. Take up the slack by advancing further into the direction of the restrictive barrier

Glenohumeral BLT/MFR

Patient seated or supine, if performed seated patient may support elbow with their opposite hand.

1. Use same grips as for segmental diagnosis
2. Place the humeral head into its motions of ease to balanced tension



3. Can also be performed as a long lever technique by moving distal humerus as described above for Dx

1st rib FPR

- Pt is supine- stand facing them- place monitoring hand (the one nearest patient) on the posterior first rib to monitor
- With the opposite hand the operator grasps the elbow, and abduct the arm to 90 degrees with full elbow flexion
- Create a compressive force through the elbow downward to the monitoring finger
- Using your monitoring arm as a fulcrum increase shoulder flexion- hold for 3-5 seconds
- Maintaining the compressive force and IR, adduct upper arm across the patient's chest and swing it down into a neutral position

Counterstrain Tenderpoints

- Bicep LH- at the long head tendon in the groove
 - flexion, and IR/ER of the humerus
- Bicep SH- Lateral aspect of the Coracoid process
 - Flexion, adduction, IR/ER
- Supraspinatus- over the supraspinatus
 - Moderate flexion, abduction and external rotation of the arm

Elbow

- **Elbow Wobble (Articulatory)**
 - Patient is supine and shoulder abducted to 45 degrees. Place the hand midway between supination and pronation (you can then pin it at your side). Grasp the elbow with both hands
 - Operator produces a wobble in a medial/lateral direction while flexing and extending in order to produce articulation at the elbow
- **Abducted/adducted elbow**
 - Same positing as above-

Clavicle treatment- BLT

- The patient is seated- stand in a split stance in front of the patient with one hand grasping the distal clavicle and one hand grasping the proximal clavicle
- Have the patient lean forward until you feel that you can "float" the clavicle
- Modify your pressure to lift either end up or down, IR or ER, compression or distraction until you reach a balance point of even tension
- Maintain this hold until you feel a release. You can fine tune by having the patient very slowly turning their head to the left or to the right
- After the release- support the clavicle as the patient sits back up- releasing only after they are back in a normal seated posture